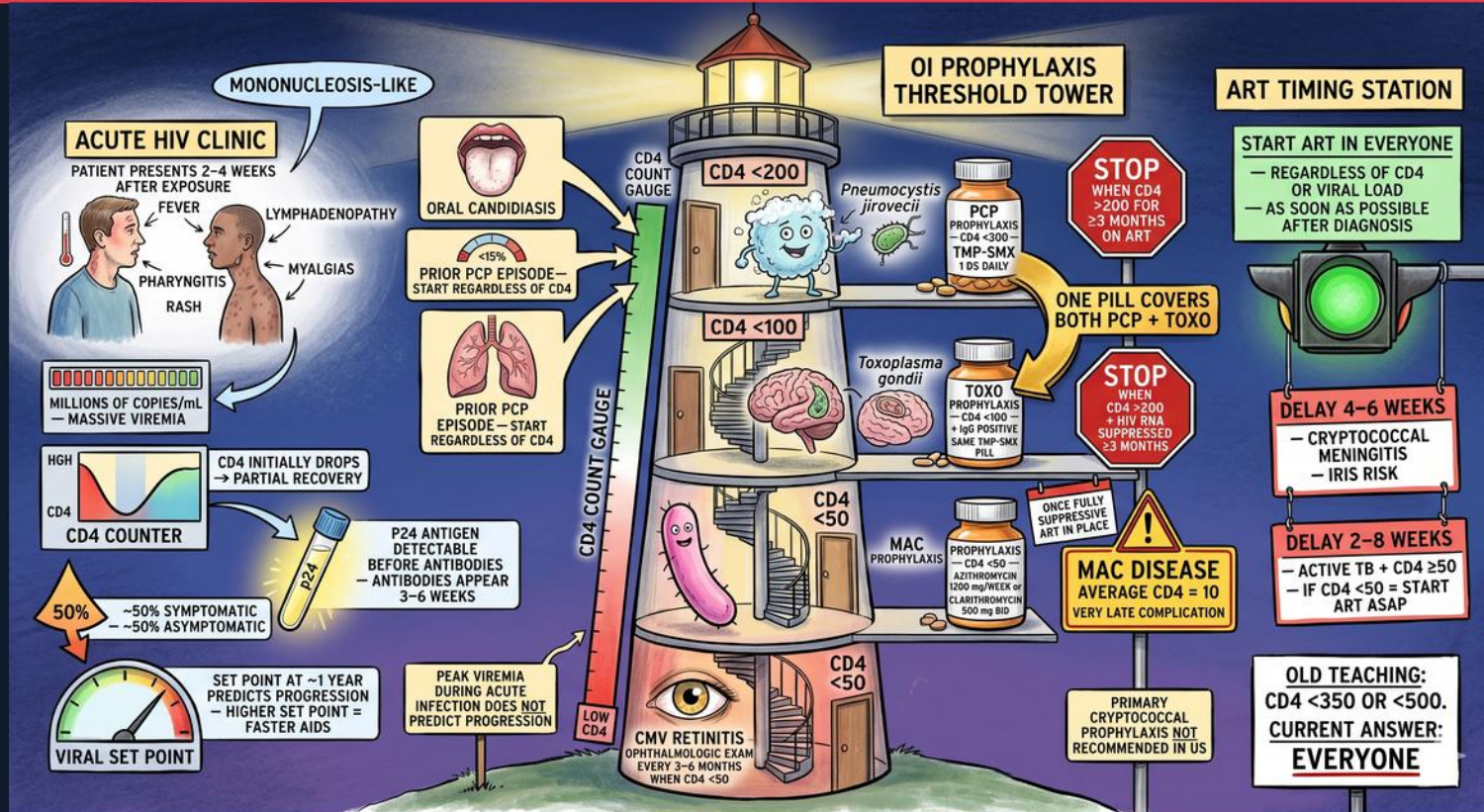
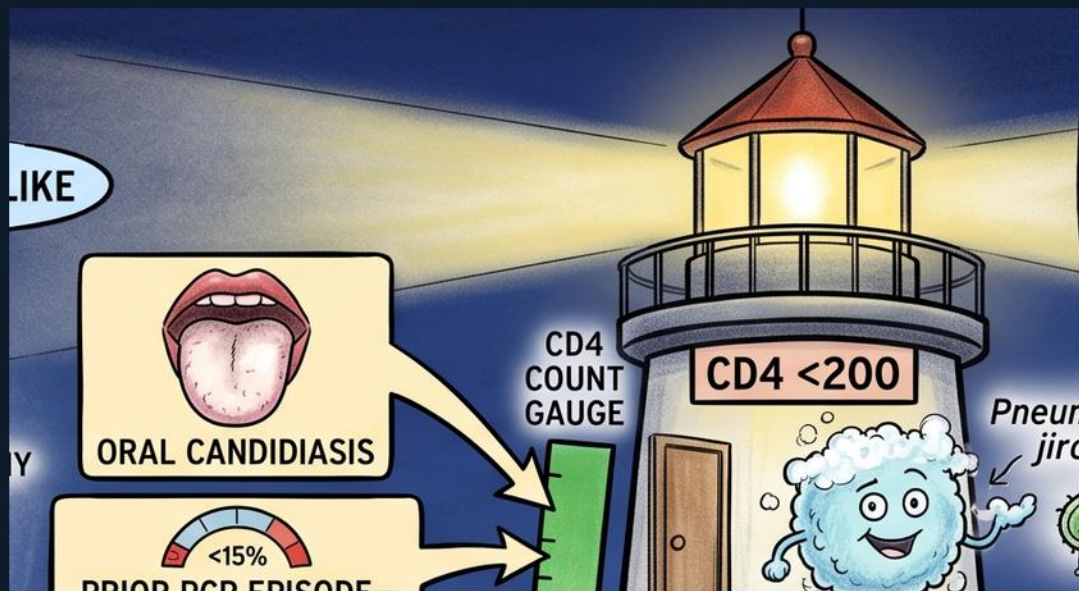


HIV — OI Prophylaxis CD4 Thresholds & ART Timing





OI Prophylaxis Threshold Tower

WHAT YOU SEE

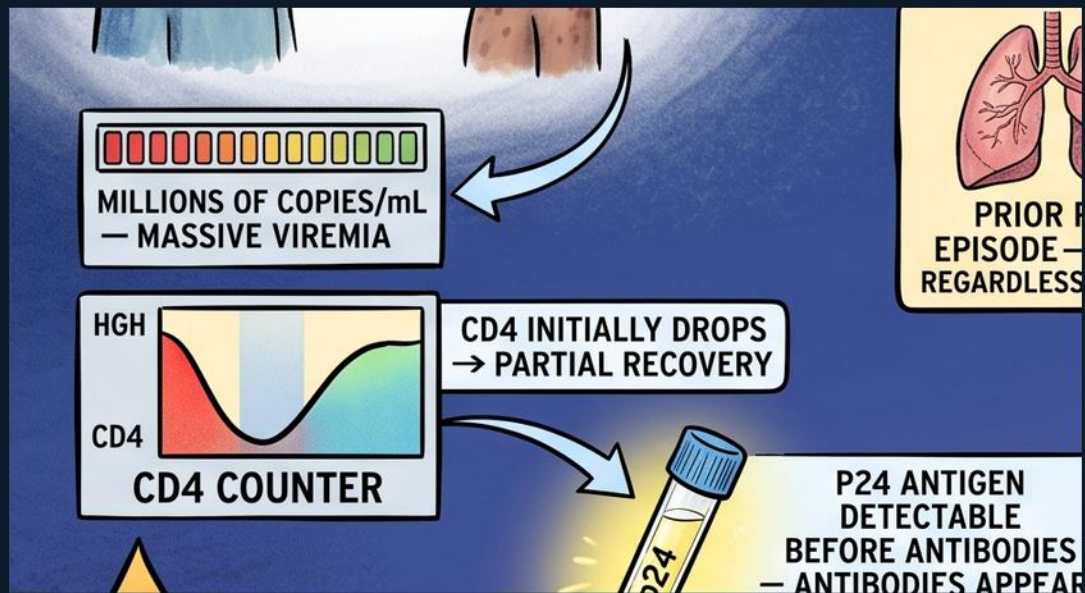
central tower labeled OI prophylaxis threshold tower with CD4 floors

WHAT IT ENCODES

Opportunistic-infection (OI) primary prophylaxis is layered by descending CD4 count, each 'floor' adding a new bug. The three numbers to memorize: CD4 <200 → start PJP prophylaxis (TMP-SMX); CD4 <100 → add Toxoplasma prophylaxis if Toxo IgG-positive (TMP-SMX covers both); CD4 <50 → historically MAC prophylaxis (azithromycin). Mnemonic: 'Pee (200) Toe (100) MAC (50)' descending — Pneumocystis, Toxo, MAC.

THE BOARD TRAP

WRONG: 'Use viral load to decide OI prophylaxis.' Prophylaxis thresholds are driven by the CD4 count, not viral load — memorize 200/100/50. Viral load guides treatment response and U=U, but the OI-risk gauge is CD4. Picking a viral-load cutoff for starting TMP-SMX is the distractor.



CD4 gauge / counter

WHAT YOU SEE

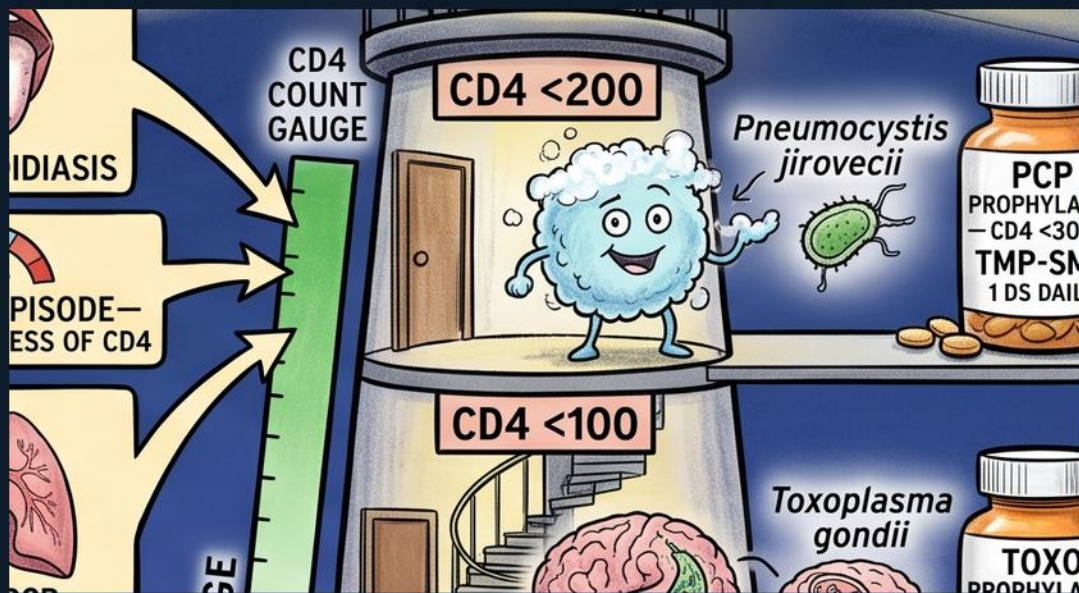
falling CD4 gauge, viral set point

WHAT IT ENCODES

The absolute CD4 count is the central driver of OI risk and the trigger for each prophylaxis layer. Normal CD4 is ~500–1500 cells/ μ L; AIDS is defined by CD4 <200 (or an AIDS-defining illness). The CD4 percentage (<14%) approximates CD4 <200 when the absolute count is borderline. Trend matters: a falling count predicts impending OI risk and need to escalate prophylaxis.

THE BOARD TRAP

WRONG: 'A drop in CD4 during acute illness means the patient now has AIDS.' Acute infections, acute HIV seroconversion, or steroids can transiently depress CD4, with partial recovery afterward — don't label AIDS off a single low value during intercurrent illness. Repeat the count when clinically stable before committing to staging/prophylaxis changes.



PJP prophylaxis — CD4 <200

WHAT YOU SEE

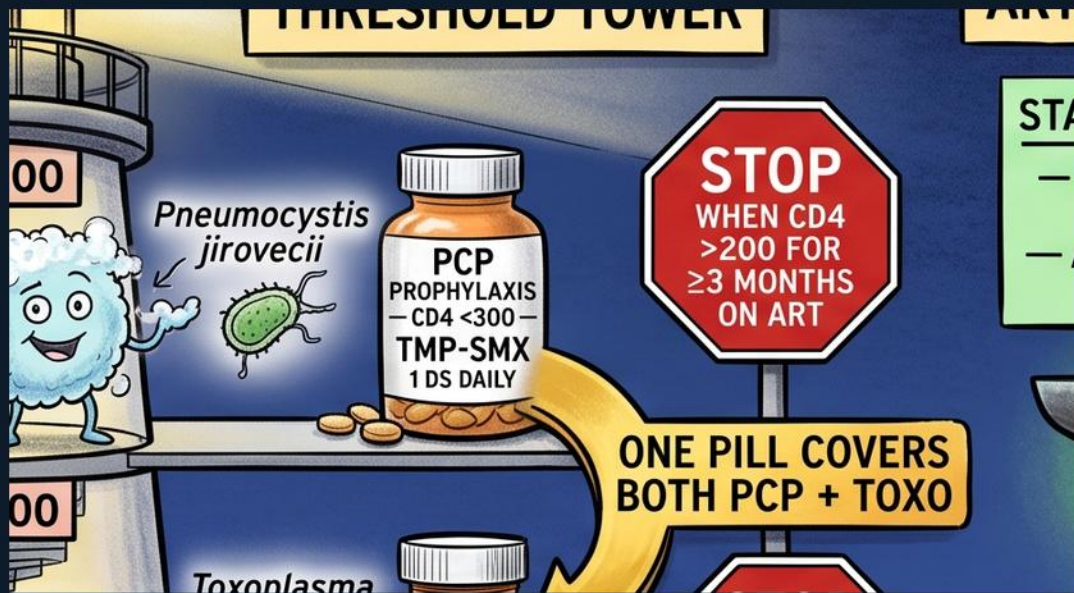
Pneumocystis lung, TMP-SMX bottle, CD4 <200 floor

WHAT IT ENCODES

Start PJP (Pneumocystis) primary prophylaxis when CD4 <200, with TMP-SMX one DS tablet daily (SS daily or DS three times weekly also effective). Also start regardless of count if there is oropharyngeal candidiasis (thrush), an AIDS-defining illness, or CD4 <14%. Alternatives for sulfa intolerance: dapsone (check G6PD), atovaquone, or aerosolized pentamidine.

THE BOARD TRAP

WRONG: 'Only the absolute CD4 <200 triggers PJP prophylaxis.' You also start if oropharyngeal candidiasis is present or there's an AIDS-defining illness, even with CD4 just above 200 — thrush is a clinical marker of risk. Anchoring solely on the number and ignoring thrush is the trap.



STOP when CD4 >200 for 3 months

WHAT YOU SEE

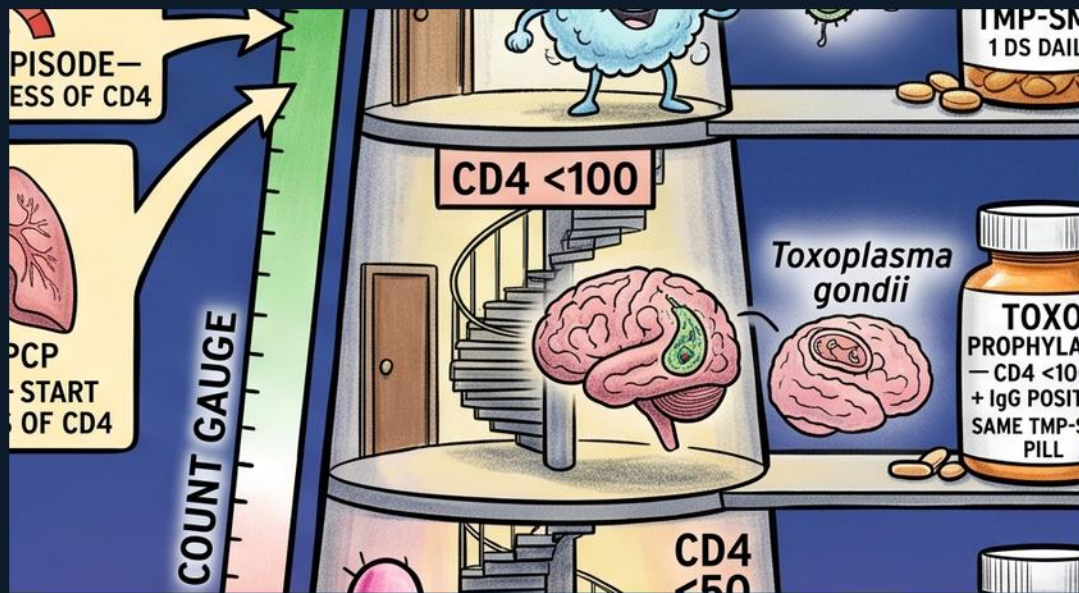
red STOP sign, CD4 >200 for ≥3 months on ART

WHAT IT ENCODES

Discontinue PJP (and Toxo) primary prophylaxis once CD4 is sustained >200 for ≥3 months on suppressive ART. Some guidance permits stopping at CD4 100–200 if the viral load is suppressed. Restart prophylaxis if CD4 falls back below 200 (or 100–200 with detectable viral load).

THE BOARD TRAP

WRONG: 'Stop prophylaxis as soon as one CD4 reads >200.' A single >200 value is not enough — you need a sustained recovery (>200 for ≥3 months) on suppressive ART. Stopping prematurely on a transient bump risks breakthrough PCP. Durability of response, not a one-off lab, is the discriminator.



Toxoplasma prophylaxis — CD4 <100

WHAT YOU SEE

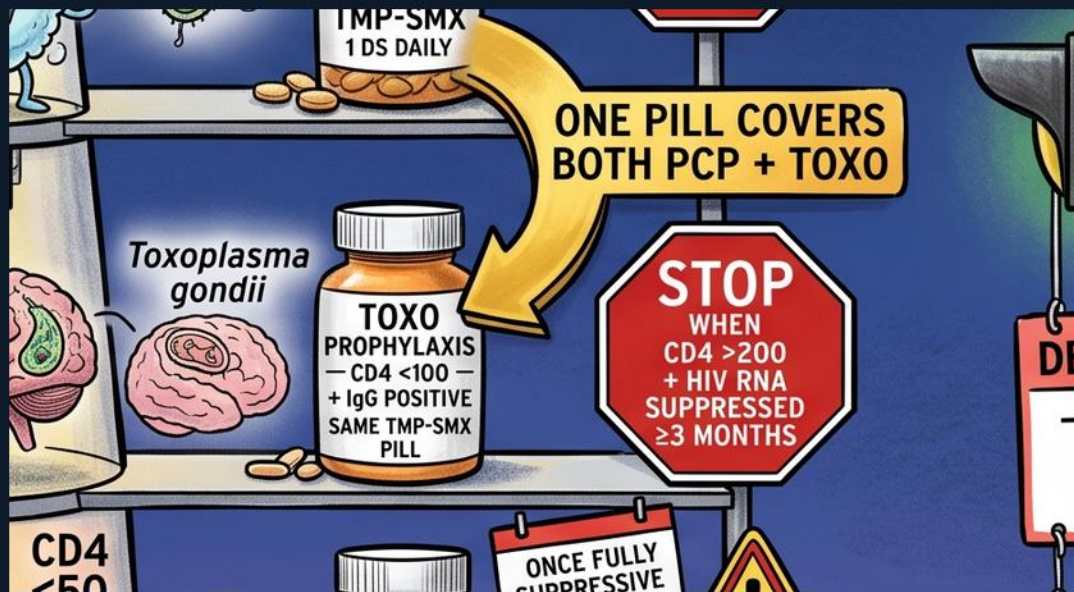
Toxoplasma cyst, CD4 <100 floor

WHAT IT ENCODES

Add Toxoplasma gondii encephalitis prophylaxis when CD4 <100 AND the patient is Toxoplasma IgG-positive (seropositive). First-line is TMP-SMX one DS daily, which conveniently also covers PJP. Toxo-seronegative patients with CD4 <100 do not need primary Toxo prophylaxis but should be counseled on avoiding exposure (undercooked meat, cat litter).

THE BOARD TRAP

WRONG: 'Give Toxo prophylaxis to every patient with CD4 <100.' Only Toxo IgG-POSITIVE patients need primary Toxo prophylaxis — a seronegative patient isn't at risk of reactivation. Forgetting to check (or ignoring) the IgG serostatus is the trap; serology gates the indication.



One pill covers PJP + Toxo

WHAT YOU SEE

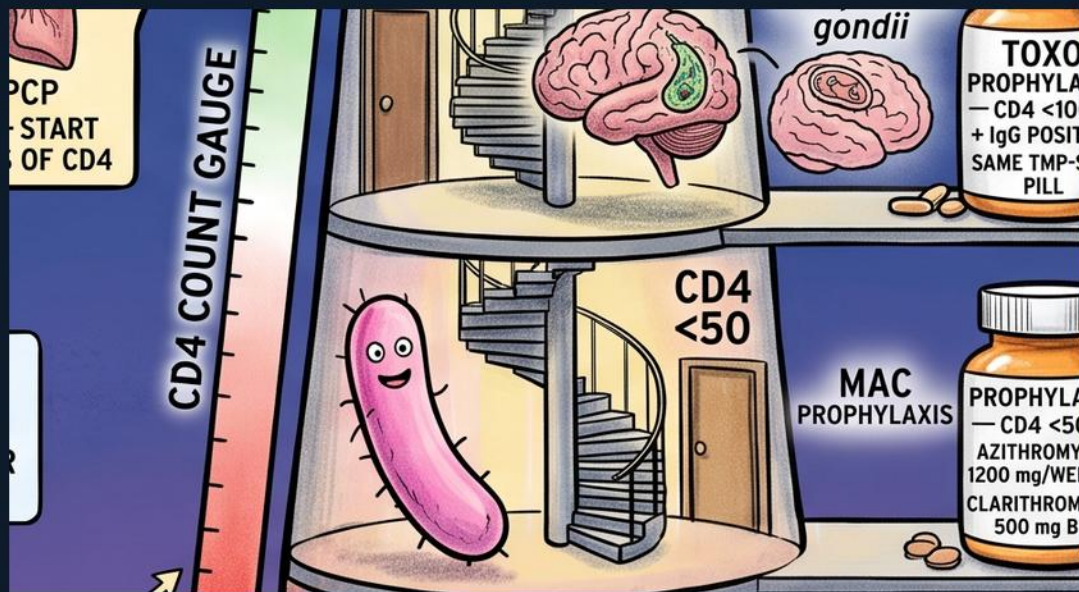
callout: one pill covers both PCP + Toxo

WHAT IT ENCODES

TMP-SMX is first-line for BOTH PJP and Toxoplasma prophylaxis — one drug, two bugs — which is why a patient at CD4 <100 on daily DS TMP-SMX is already covered for both. This is a high-yield efficiency point on exams. If the patient cannot take TMP-SMX, the alternative regimens must individually account for Toxo coverage.

THE BOARD TRAP

WRONG: 'Any sulfa-allergy alternative for PJP also covers Toxo.' Alternatives differ: dapsone-pyrimethamine-leucovorin and atovaquone cover Toxo, but aerosolized pentamidine and dapsone-monotherapy do NOT cover Toxoplasma. Always confirm G6PD before dapsone/primaquine. Assuming the PJP substitute automatically covers Toxo is the trap.



MAC prophylaxis — CD4 <50

WHAT YOU SEE

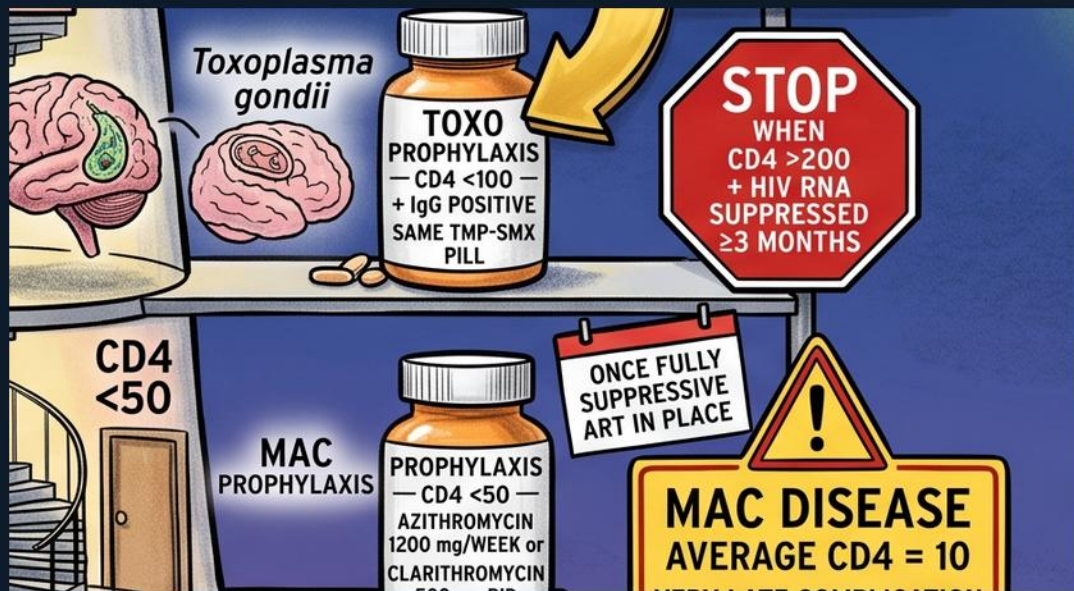
MAC bacteria, CD4 <50 floor

WHAT IT ENCODES

Historically, MAC primary prophylaxis (azithromycin 1200 mg weekly, or clarithromycin) was started at CD4 <50. Current guidance: routine primary MAC prophylaxis is **NO LONGER** recommended for patients who start effective ART promptly, because ART-driven immune recovery prevents disseminated MAC. It is reserved for those not on/responding to ART. Rule out active MAC disease before starting prophylaxis.

THE BOARD TRAP

WRONG: 'Always give azithromycin MAC prophylaxis at CD4 <50.' Updated guidance dropped routine primary MAC prophylaxis when ART is started promptly — ART itself prevents MAC. Reflexively starting azithromycin for every CD4 <50 patient who is beginning ART is now the outdated/trap answer.



MAC average CD4 ~10

WHAT YOU SEE

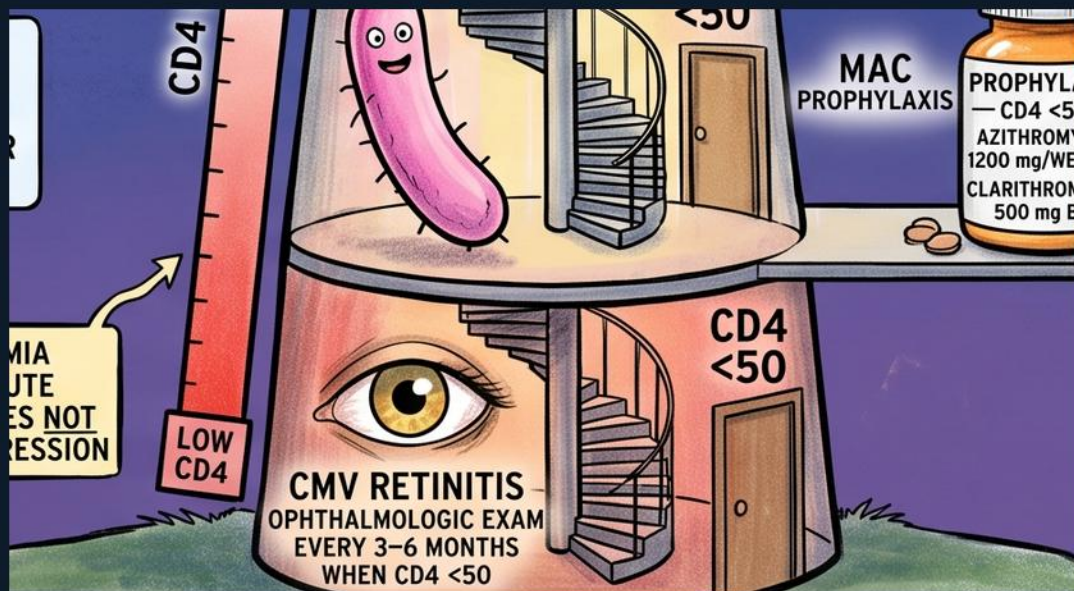
warning triangle, MAC disease average CD4 ~10, very late

WHAT IT ENCODES

Disseminated MAC is an end-stage OI — the average CD4 at disease onset is around 10. It signals profound immunosuppression and is associated with persistent fevers, night sweats, weight loss, anemia, elevated alkaline phosphatase, and hepatosplenomegaly. Because it is so late, the modern strategy is prompt ART (which prevents it) rather than primary chemoprophylaxis.

THE BOARD TRAP

WRONG: 'MAC can occur at any CD4 like TB.' MAC disseminated disease is specifically a very-late/end-stage event (average CD4 ~10), whereas TB can occur at any CD4. The discriminator: extremely low CD4, blood-culture positivity, high alk phos, and hepatosplenomegaly point to MAC. Treating it as an early/any-CD4 problem is wrong.



CMV retinitis — CD4 <50

WHAT YOU SEE

eye/CMV retinitis at CD4 <50 floor

WHAT IT ENCODES

CMV end-organ disease (especially retinitis) occurs at CD4 <50, presenting with painless floaters, blurred vision, or visual field loss and a 'pizza-pie' (hemorrhage + exudate) fundus. There is NO primary CMV prophylaxis; management is dilated-eye surveillance in advanced disease plus treatment when disease appears (valganciclovir oral, or ganciclovir/foscarnet IV; intravitreal therapy for sight-threatening lesions).

THE BOARD TRAP

WRONG: 'Start prophylactic valganciclovir at CD4 <50 to prevent CMV.' Primary CMV prophylaxis is NOT recommended (toxicity, resistance, no mortality benefit) — instead use surveillance and treat established disease. The discriminator from MAC/Toxo: CMV has no primary prophylaxis tier, only screening and treatment.



Start ART in EVERYONE

WHAT YOU SEE

green light station, start ART in everyone regardless of CD4

WHAT IT ENCODES

Treat ALL people with HIV with ART regardless of CD4 count or viral load, and start as soon as possible after diagnosis (even same-day in many programs). Benefits: lower mortality, slower progression, immune preservation, and prevention of transmission (U=U). Preferred initial regimens are integrase-inhibitor based, e.g., bicitgravir/tenofovir alafenamide/emtricitabine or dolutegravir-based combinations.

THE BOARD TRAP

WRONG: 'Wait until CD4 falls below 350 (or 500) before starting ART.' The old CD4-threshold-to-treat teaching is obsolete — the current answer is treat everyone, immediately, regardless of CD4 or viral load. Picking a CD4 cutoff to initiate ART is the classic outdated trap.



Delay 4–6 weeks — cryptococcal meningitis

WHAT YOU SEE

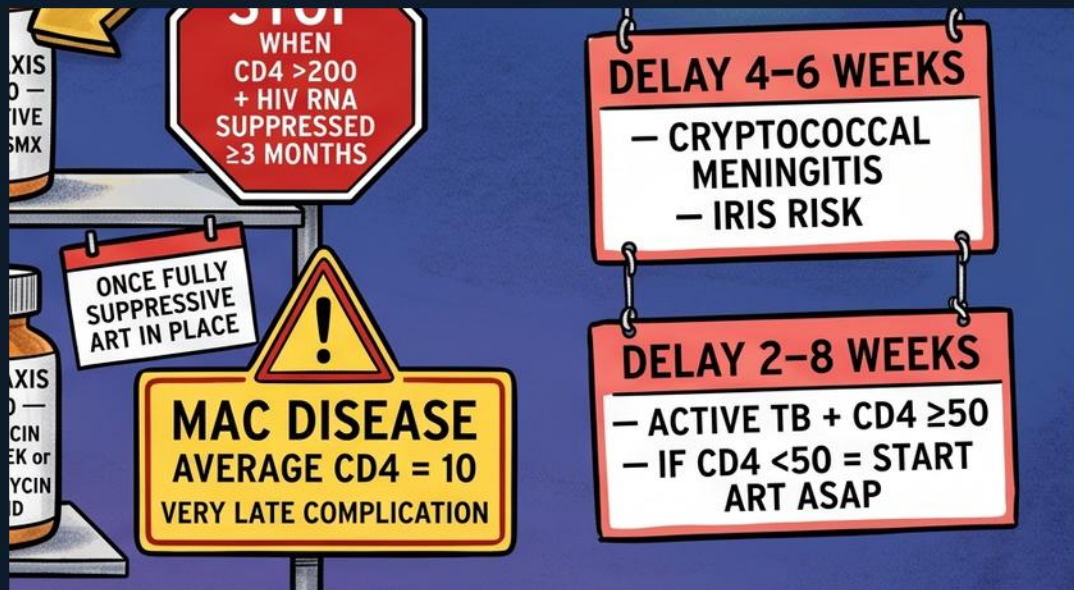
delay 4–6 weeks box, cryptococcal meningitis, IRIS

WHAT IT ENCODES

Cryptococcal meningitis is the major EXCEPTION to 'start ART immediately': delay ART ~4–6 weeks after beginning antifungal therapy. Early ART here triggers immune reconstitution inflammatory syndrome (IRIS) within the CNS, which significantly increases mortality. First control the infection (ampho B + flucytosine induction, manage raised ICP), then introduce ART.

THE BOARD TRAP

WRONG: 'Cryptococcal meningitis is treated like any OI — start ART right away.' Concurrent/early ART in crypto meningitis increases death from CNS IRIS; you must delay ~4–6 weeks. This is the standout exception to the 'treat all promptly' rule, and choosing immediate ART is the trap.



Delay 2–8 weeks — TB

WHAT YOU SEE

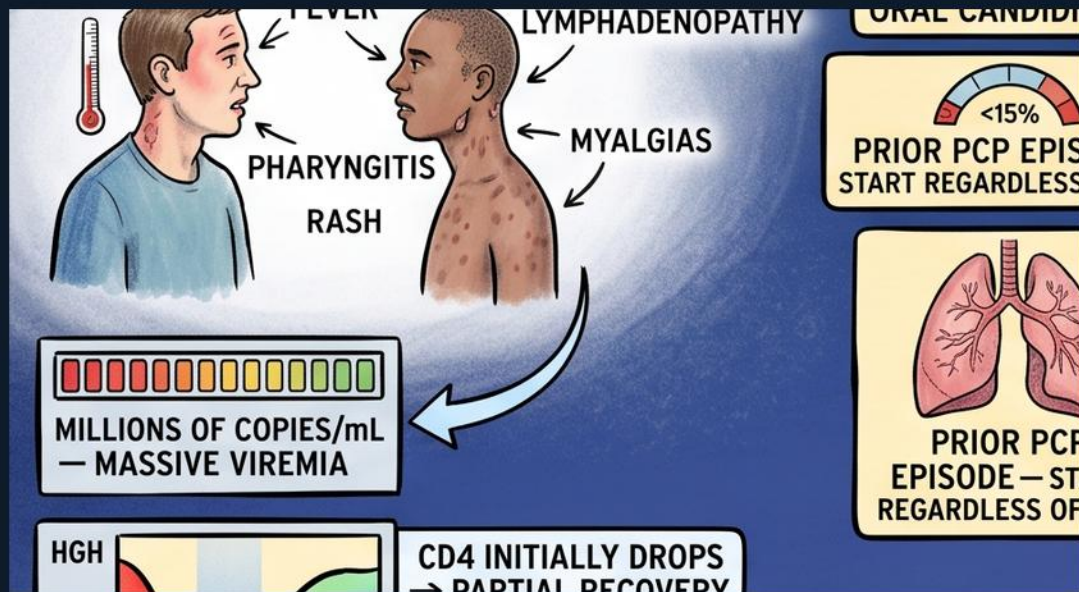
delay 2–8 weeks box, active TB CD4 timing

WHAT IT ENCODES

In active TB, start anti-TB therapy first, then add ART with timing based on CD4: if CD4 <50, start ART within 2 weeks of TB treatment; if CD4 ≥50, start by ~8 weeks. This balances the survival benefit of early ART against TB-IRIS risk. (TB meningitis is handled more cautiously given the danger of CNS IRIS.)

THE BOARD TRAP

WRONG: 'Start ART and TB drugs simultaneously in everyone with TB.' ART timing is CD4-dependent — within 2 weeks only if CD4 <50, otherwise around 8 weeks — to mitigate TB-IRIS. Also watch rifampin–ART interactions (rifabutin or dose adjustments). Ignoring CD4-based timing and rifampin interactions is the trap.



P24 antigen / window period

WHAT YOU SEE

p24 antigen detectable before antibodies, 3–6 weeks

WHAT IT ENCODES

During acute HIV, p24 antigen and HIV RNA become detectable BEFORE antibodies seroconvert (~3–6 weeks; the 'window period'). Acute retroviral syndrome looks mononucleosis-like (fever, pharyngitis, lymphadenopathy, rash, myalgias) with very high viral load. Fourth-generation antigen/antibody combo assays detect p24 + antibodies, narrowing the window; HIV RNA is earliest.

THE BOARD TRAP

WRONG: 'A negative HIV antibody test excludes acute HIV.' Antibody-only tests miss acute/early HIV in the window period — use a 4th-generation Ag/Ab combo assay or HIV RNA when acute infection is suspected. A patient with mono-like illness and high-risk exposure but negative antibody needs RNA/p24 testing, not reassurance.

Summary — every symbol

1. OI Prophylaxis Threshold Tower

Opportunistic-infection (OI) primary prophylaxis is layered by descending CD4 count, each 'floor' adding a new bug

3. PJP prophylaxis — CD4 <200

Start PJP (Pneumocystis) primary prophylaxis when CD4 <200, with TMP-SMX one DS tablet daily (SS daily or DS three times weekly also effective)

5. Toxoplasma prophylaxis — CD4 <100

Add Toxoplasma gondii encephalitis prophylaxis when CD4 <100 AND the patient is Toxoplasma IgG-positive (seropositive)

7. MAC prophylaxis — CD4 <50

Historically, MAC primary prophylaxis (azithromycin 1200 mg weekly, or clarithromycin) was started at CD4 <50

9. CMV retinitis — CD4 <50

CMV end-organ disease (especially retinitis) occurs at CD4 <50, presenting with painless floaters, blurred vision, or visual field loss and a 'pizza-pie' (hemorrhage + exudate) fundus

11. Delay 4–6 weeks — cryptococcal meningitis

Cryptococcal meningitis is the major EXCEPTION to 'start ART immediately': delay ART ~4–6 weeks after beginning antifungal therapy

13. P24 antigen / window period

During acute HIV, p24 antigen and HIV RNA become detectable BEFORE antibodies seroconvert (~3–6 weeks; the 'window period')

2. CD4 gauge / counter

The absolute CD4 count is the central driver of OI risk and the trigger for each prophylaxis layer

4. STOP when CD4 >200 for 3 months

Discontinue PJP (and Toxo) primary prophylaxis once CD4 is sustained >200 for ≥3 months on suppressive ART

6. One pill covers PJP + Toxo

TMP-SMX is first-line for BOTH PJP and Toxoplasma prophylaxis — one drug, two bugs — which is why a patient at CD4 <100 on daily DS TMP-SMX is already covered for both

8. MAC average CD4 ~10

Disseminated MAC is an end-stage OI — the average CD4 at disease onset is around 10

10. Start ART in EVERYONE

Treat ALL people with HIV with ART regardless of CD4 count or viral load, and start as soon as possible after diagnosis (even same-day in many programs)

12. Delay 2–8 weeks — TB

In active TB, start anti-TB therapy first, then add ART with timing based on CD4: if CD4 <50, start ART within 2 weeks of TB treatment; if CD4 ≥50, start by ~8 weeks